Group Portfolio

Project-management artefacts, integration diagrams, user journey maps and test evidence

Team	Distributed Developments (group project)
Members	Mohammed Nurul Hasan (S376146)
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Product	Library Platform, .NET 10 ASP.NET Core MVC
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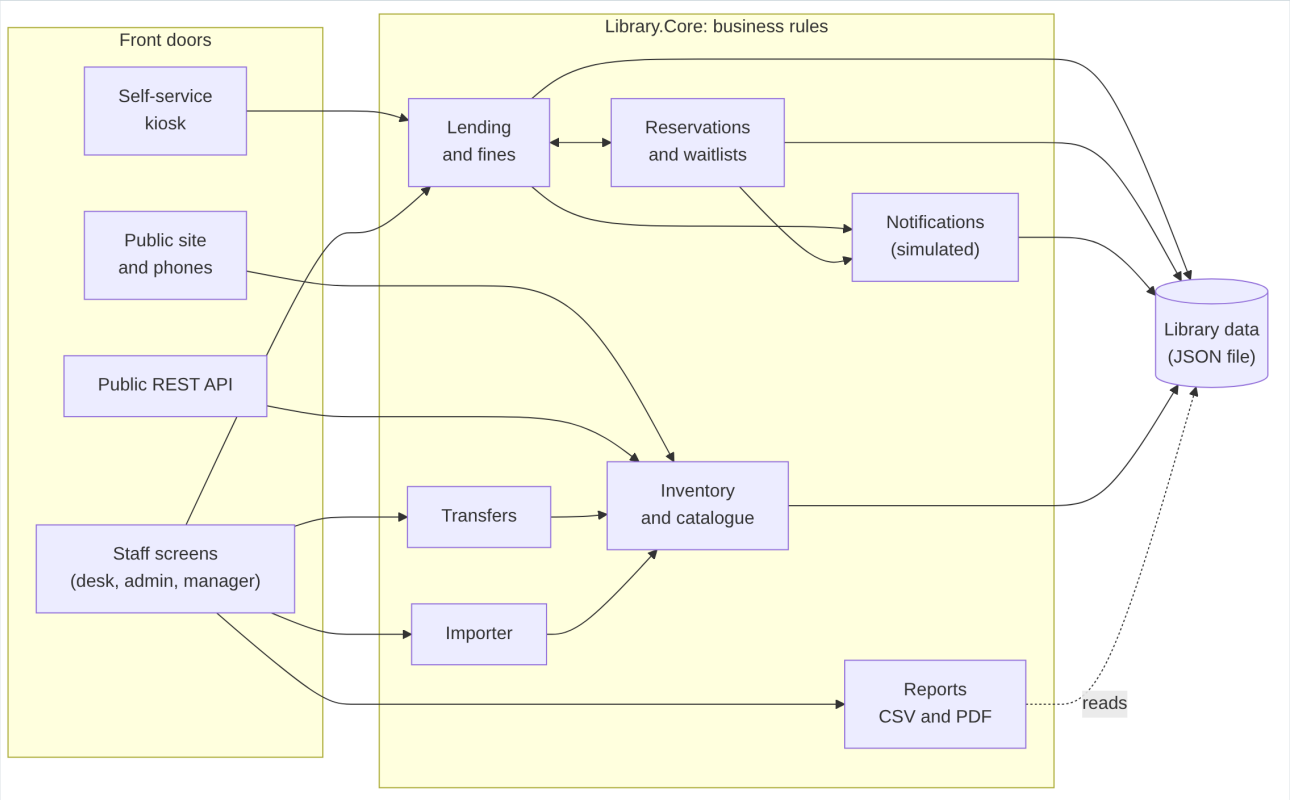
This portfolio collects the project-management artefacts, the integration diagrams and the user journey maps for the Library Platform, with the evidence that the integrated product works.

Team member	Student ID
Mohammed Nurul Hasan	S376146
Md Rakibul Hassan Emon	S375332
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The code, README and guides are in the repository <https://github.com/nurulhasan2k/library-platform.git>.

1. Project at a glance

The client asked for the Assignment 2 library system to be expanded into **one coherent platform** rather than seven prototypes. The result is a single .NET 10 ASP.NET Core MVC application in which a patron can search from a phone, borrow at a kiosk or a desk, join a waiting list, be notified; staff can move items between branches and import stock; and managers can see and export reports, all from the same data.



2. Project-management artefacts

2.1 Phases and what each produced

Phase	Focus	Result
0 Kickoff	Agree the contract and split the work	Repository with standards and PR template; service contracts and stubs; ER diagram; seed scenarios; 14 design decisions; project board with 68 cards
1 Foundation	Stabilise the baseline and the multi-branch data model	Assignment 2 rules ported to .NET 10 MVC; roles and sign-in; branches, desks, categories; branch-aware lending; core seed data

Phase	Focus	Result
2 Features	Build the seven features in parallel	Kiosk, API, notifications, transfers, reservations, importer, analytics
3 Integration	Connect everything and run the demo list	Seven integration chains tested end to end; 12 demo items checked
4 Documentation	Guides and evidence	README (14 topics), installation guide, user guide, plan, portfolio

2.2 Ownership (who is responsible and who supplies)

R = responsible (leads), S = supplies a part, V = reviews.

Area	Mohammed Nurul Hasan	Md Rakibul Hassan Emon	Md Modasser Hossain
Foundation: MVC port, data, roles, layout	R	V	S
3.4 Multi-branch inventory and transfers	R	S	S (report)
3.1 Self-service kiosk	S (lending)	R	V
3.3 Notifications	S (hooks)	R	S (counts for reports)
3.5 Reservations and waitlists	S (status hooks)	R	V
3.2 Public REST API	S (data)	V	R
3.6 Data importer	S (branches, rules)	V	R
3.7 Manager analytics and export	S (roles, history)	S (queue counts)	R
Seed and reset	R	S	S
README, Installation guide	R	S	S
User Guide, Project Plan	V	R	S
Group Portfolio	S	S	R
Code review rotation	reviews Rakibul	reviews Modasser	reviews Mohammed

2.3 Work packages

68 tasks on the project board, each with an owner, a milestone and an estimate. Totals by milestone (estimated hours):

Team member	Kickoff	Midpoint	Final	Total
Mohammed Nurul Hasan	7	78	40	125
Md Rakibul Hassan Emon	6	59	64	129
Md Modasser Hossain	4	63	70	137
Team	17	200	174	391

The full task list is Appendix A of the Project Plan, and the Plan's section 7.1 records who committed what, from the Git history.

2.4 Risk register

#	Risk	Likelihood	Impact	Mitigation	Outcome
R1	Three people build incompatible parts	High	High	Contract-first: interfaces, data model, roles matrix and seed data agreed before building; stubs so everything compiles	Contracts kept; stubs replaced by real services
R2	The integrated demo does not work on the assessor's virtual machine	Medium	High	No external packages, no database server, one command to run and one to reset; installation guide with troubleshooting; scripted fresh-install check	Fresh clone builds and runs; VM image itself still to be checked
R3	NuGet or internet is not available on the VM	Medium	High	Use only what ships with the .NET 10 SDK	Zero third-party packages
R4	Meaning of "branch" is ambiguous	Medium	Medium	Assume library branch; state it in README and decisions; ask the client	Assumption documented
R5	Time runs short (the three planned meetings were not held)	High	Medium	Decisions recorded in writing; shared project board; automated tests keep integration honest	Decisions in `docs/DECISIONS.md`; all features completed
R6	Demonstrating time-based notifications takes days	Medium	Medium	Admin "simulated today" and "Run checks now" using the same code as the daily job	Demonstrable in a minute
R7	Imported data corrupts the catalogue	Medium	High	Validate every row, preview before saving, re-check at commit, import history, problems file	12 importer tests; sample files for every rule
R8	Reports disagree between screen and exports	Medium	Medium	One report document builds the dashboard, the CSV and the PDF	Tested: dashboard figures equal the reports
R9	Public API exposes private data or is abused	Low	High	Read-only (405 otherwise), key, rate limit, DTOs without patron or loan data	Tested by API checks
R10	Demo data goes stale	Medium	Medium	Seeded dates are relative to today; `--reset-db`	Reset tested

2.5 Tools and communication

GitHub repository and GitHub Projects board for tasks; pull requests with a template and one reviewer each; a team chat with a twice-weekly written check-in; `docs/DECISIONS.md` as the decision log; the automated tests and the scripted browser run as the quality gate.

2.6 Decision log (summary)

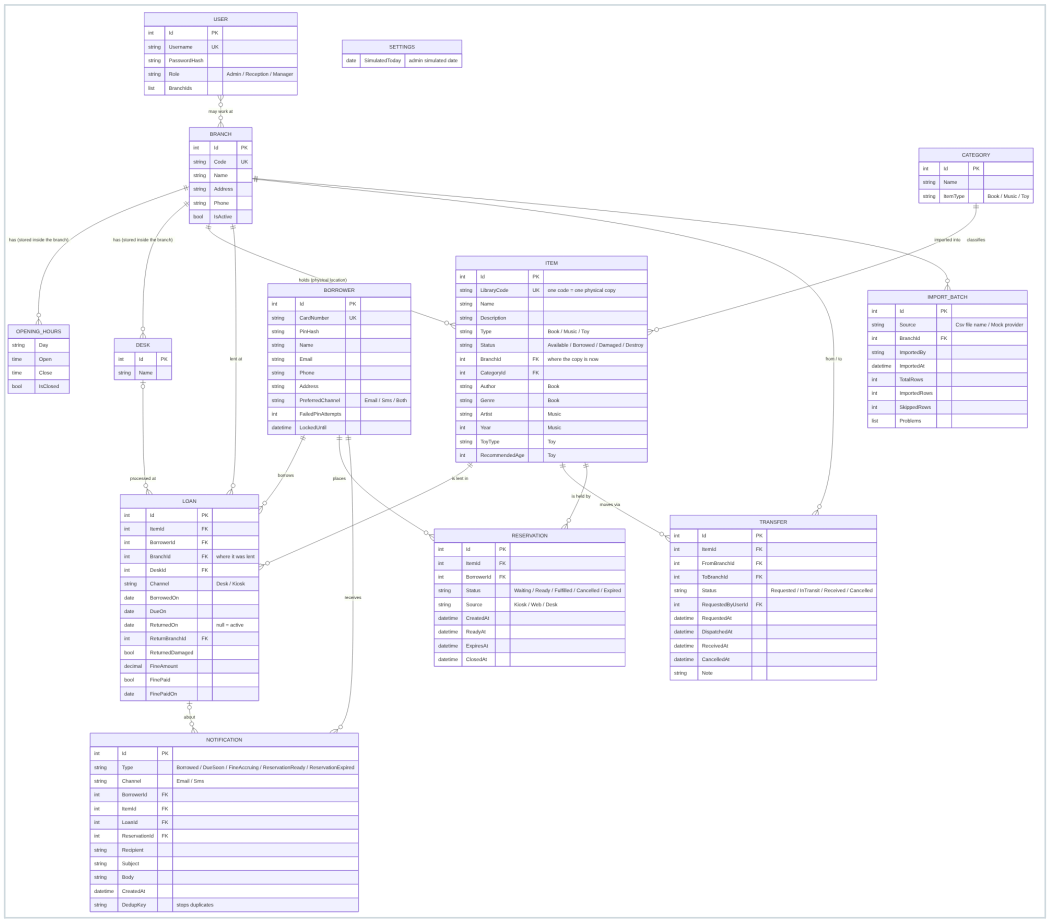
#	Decision	Chosen
D1	Data storage	One JSON file behind a storage interface (nothing to install; easy reset)
D2	Authentication	Cookie authentication, three schemes (staff, patron, kiosk), roles Admin / Reception / Manager
D3	Patron identity	Card number + 4-digit PIN, hashed, lock-out after 5 wrong tries
D4	Reservation granularity	Per physical copy
D5, D6	Where an item is, in transit	The branch holding it; "in transit" derived from an active transfer
D9, D10	Notifications and time	Simulated email/SMS; admin simulated date and "Run checks now"
D11, D12	CSV, PDF, API docs	Hand-written writers and OpenAPI document (no packages)
D13	API protection	Read-only, API key, rate limit

The full list with reasons and trade-offs is in `docs/DECISIONS.md`.

3. System diagrams

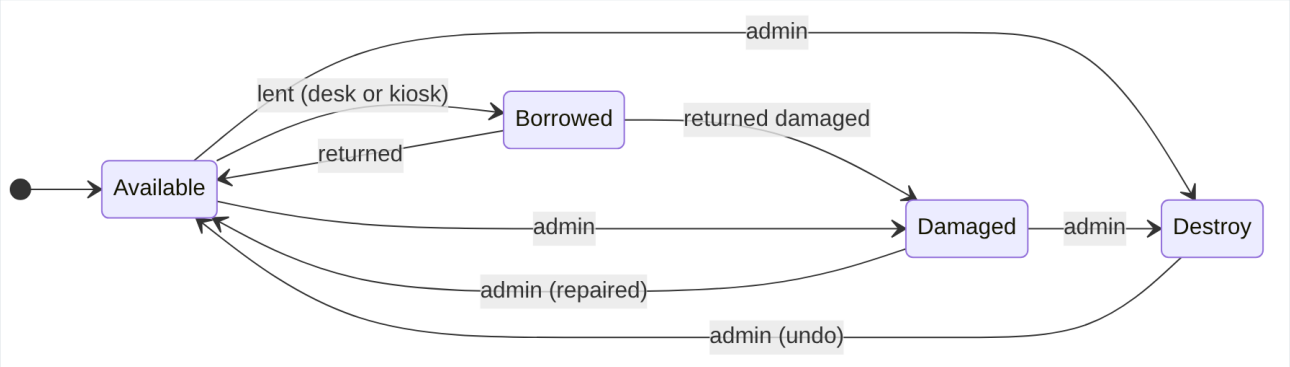
3.1 Data model

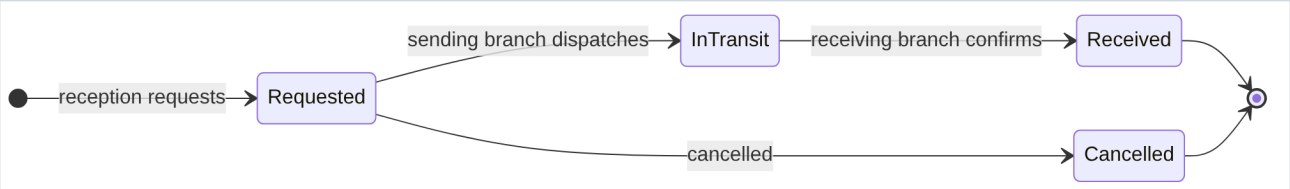
The data model (full diagram in `docs/ER-DIAGRAM.md`):



Entity-relationship diagram

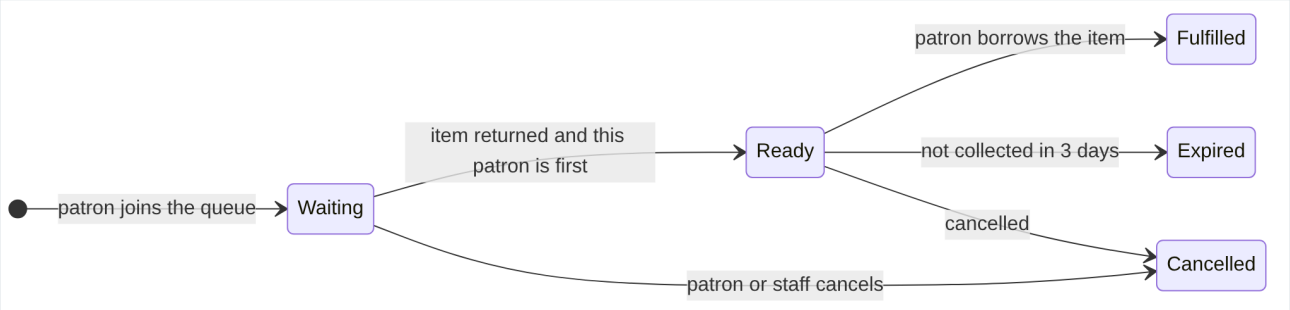
3.2 Item status and transfer life cycle





While a transfer is *Requested* or *InTransit* the item cannot be lent. The item changes branch only when the transfer is *Received*.

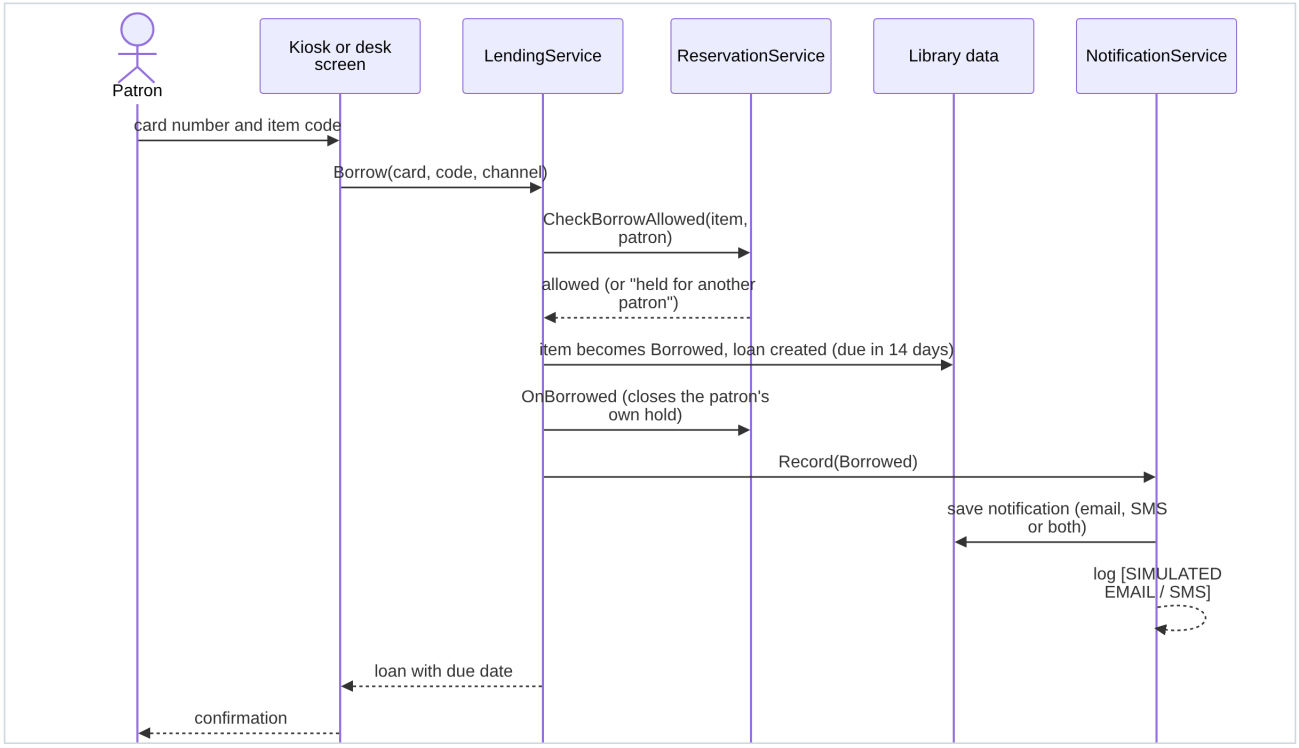
3.3 Reservation life cycle



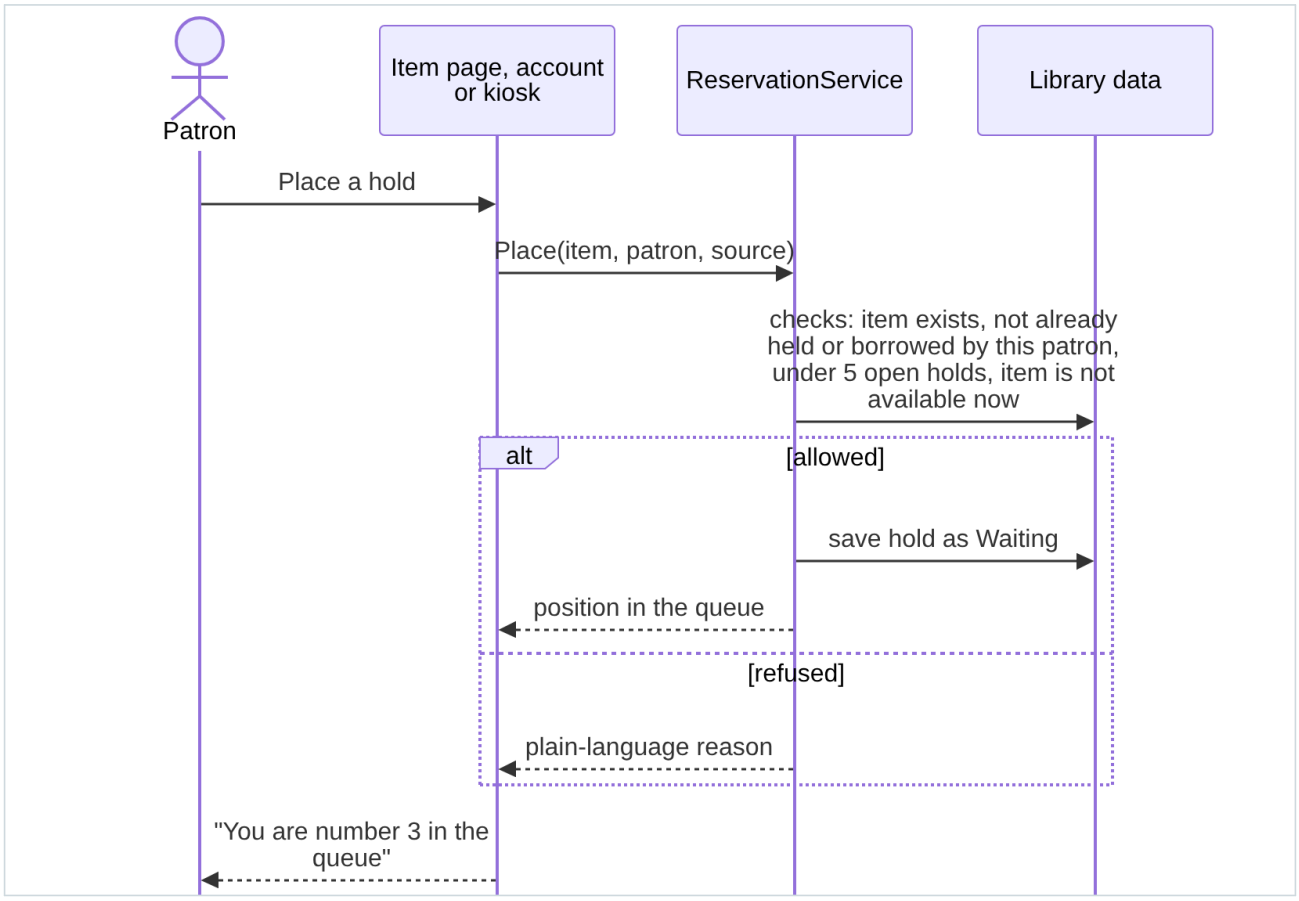
4. Integration diagrams: the seven workflows

Each diagram shows the real calls between parts of the system.

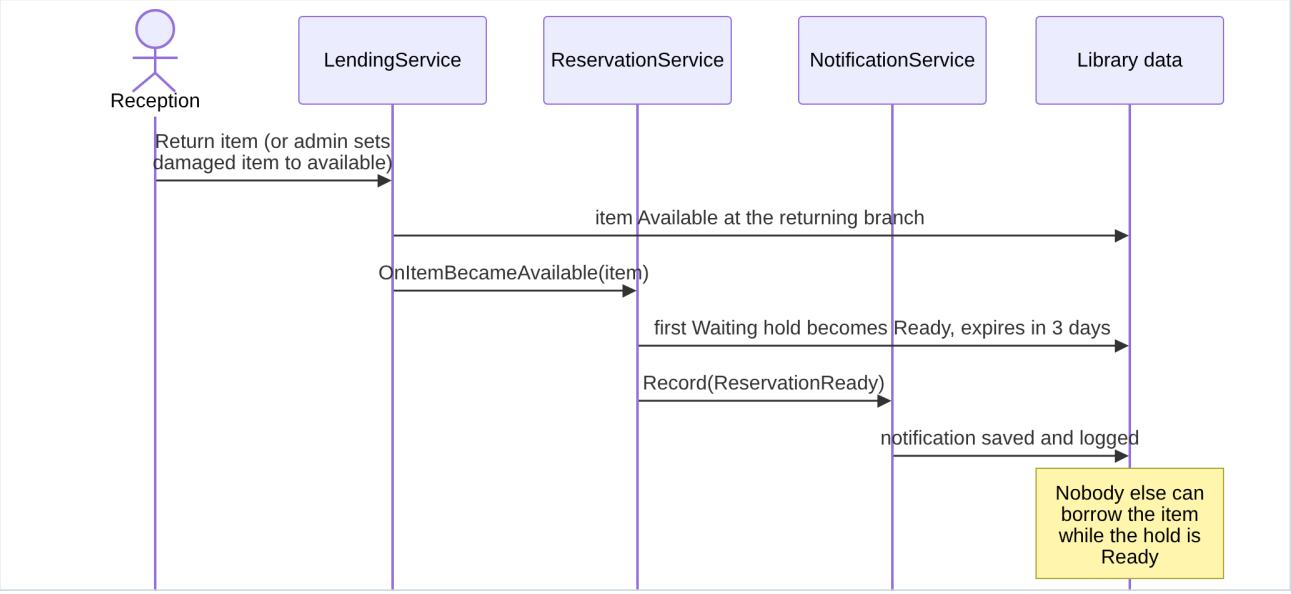
Chain 1. A patron borrows an item, the inventory is updated, a notification is recorded



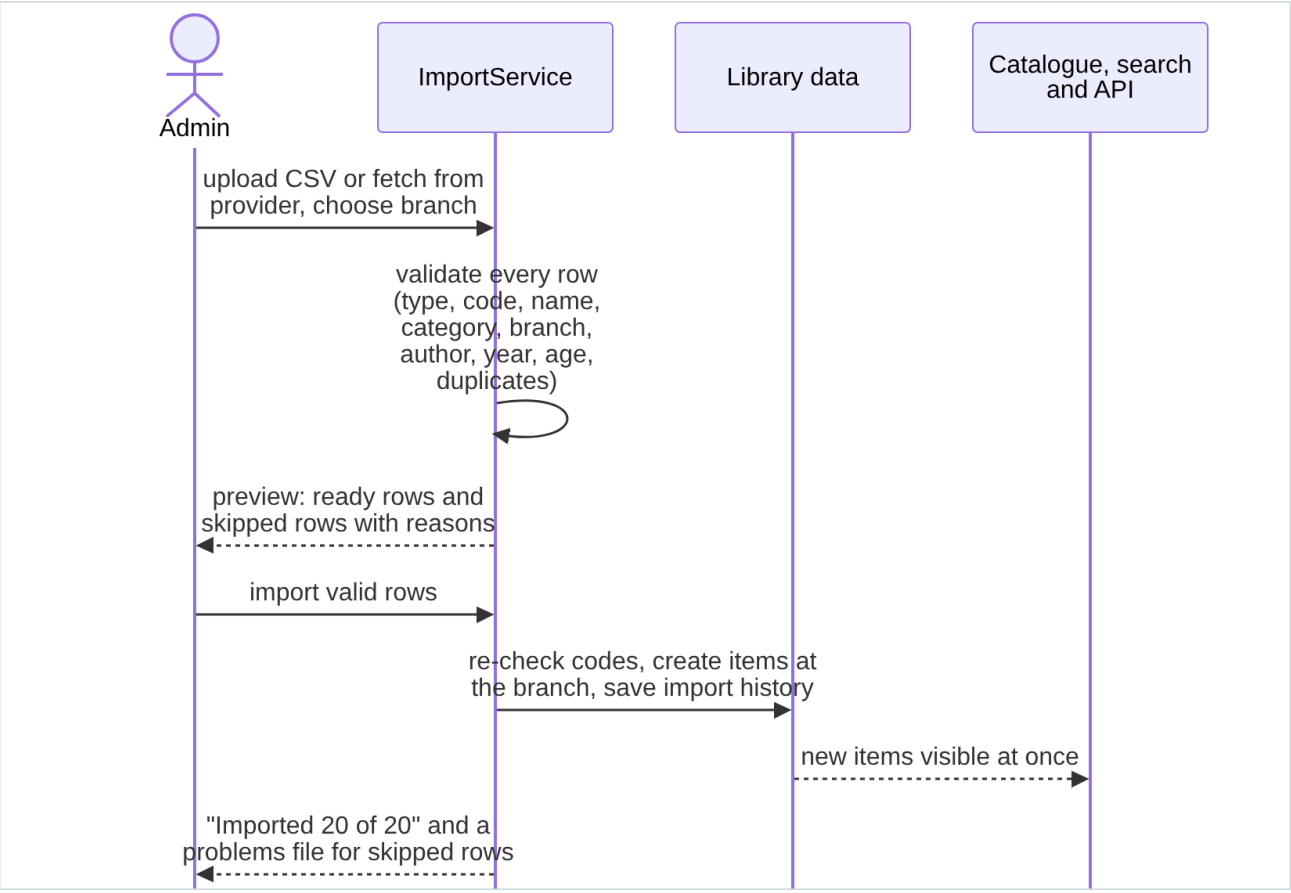
Chain 2. An item is unavailable and a patron joins its waitlist



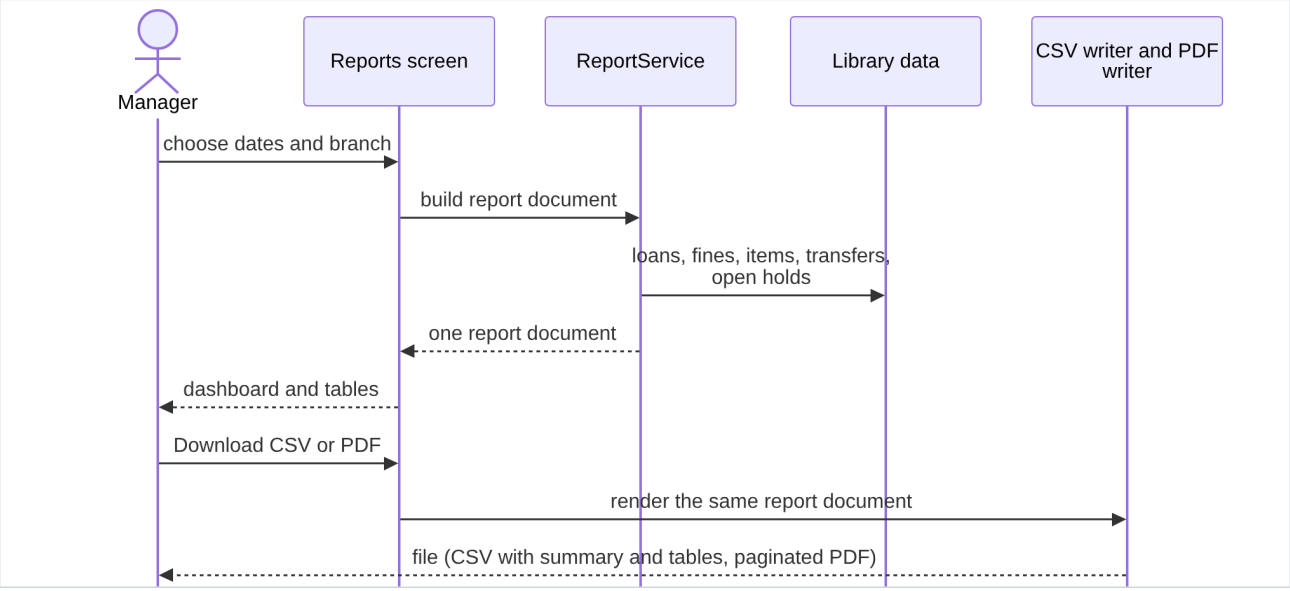
Chain 3. An item becomes available and the next patron is notified



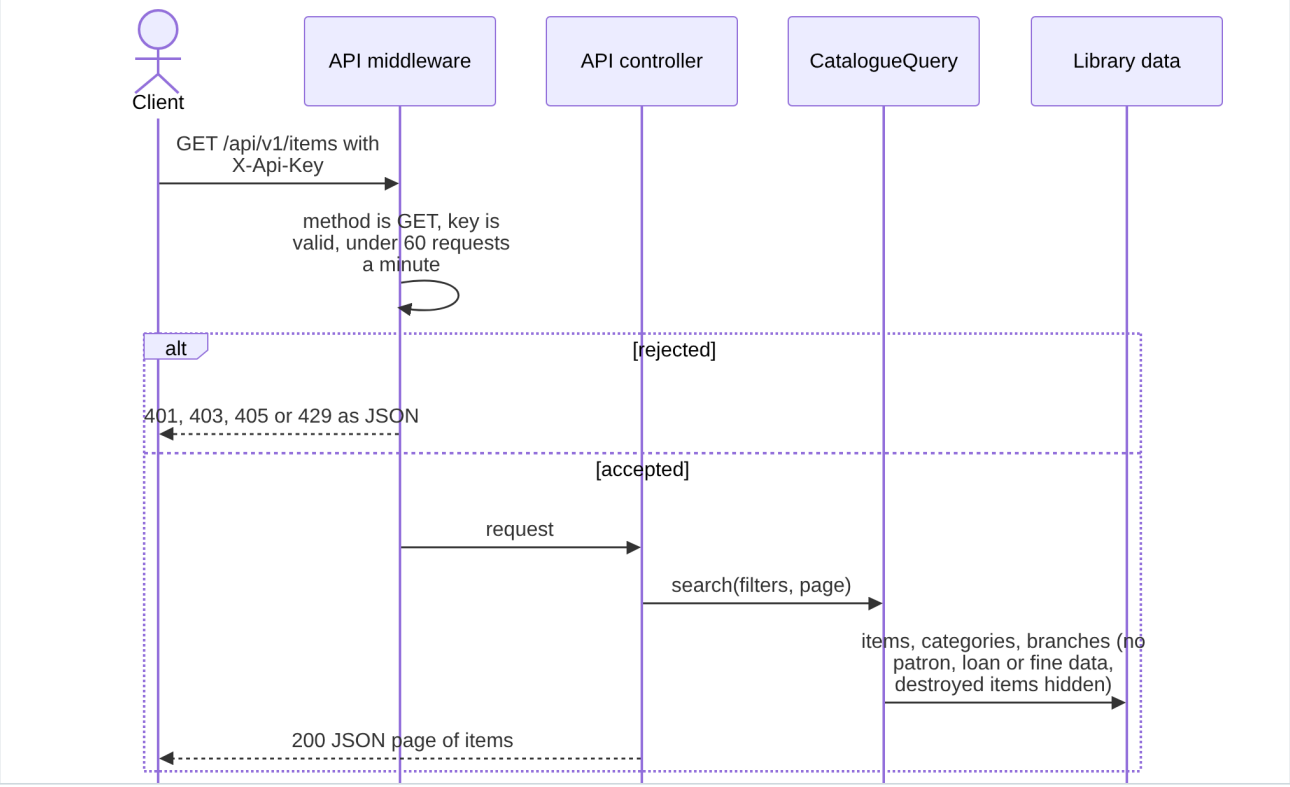
Chain 4. An administrator imports items and they appear in the catalogue



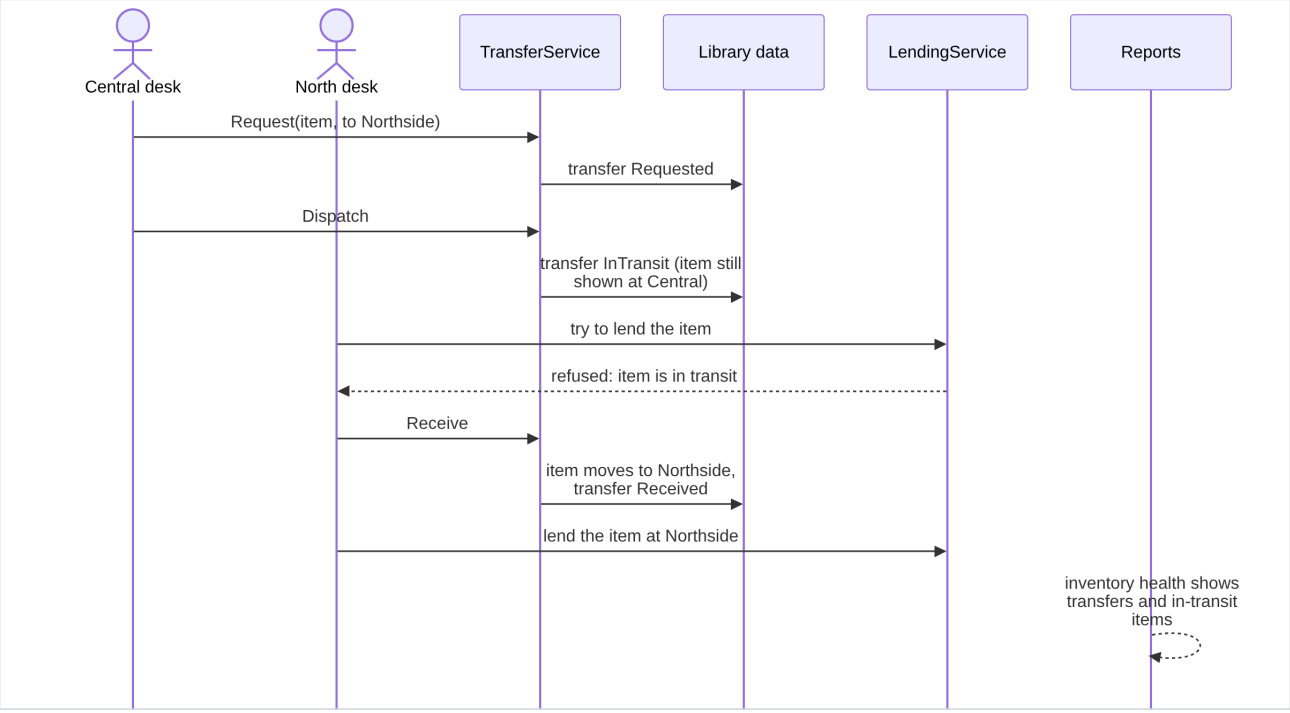
Chain 5. A manager views reports and exports them



Chain 6. An external client queries the catalogue through the API



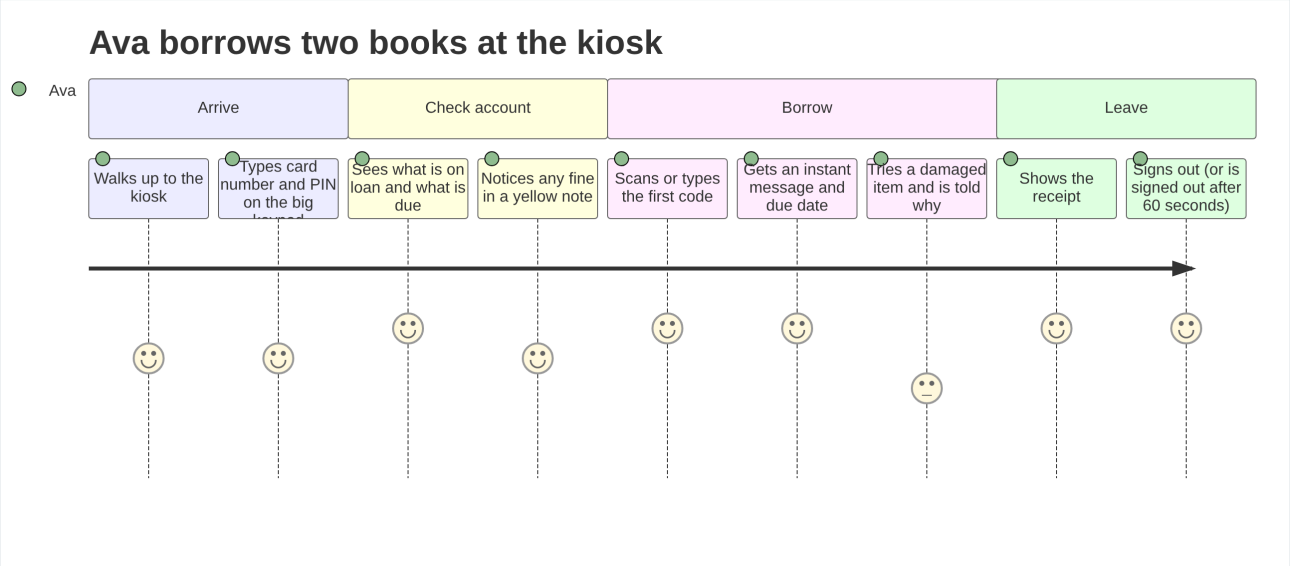
Chain 7. Staff identify item branches and coordinate transfers



5. User journey maps

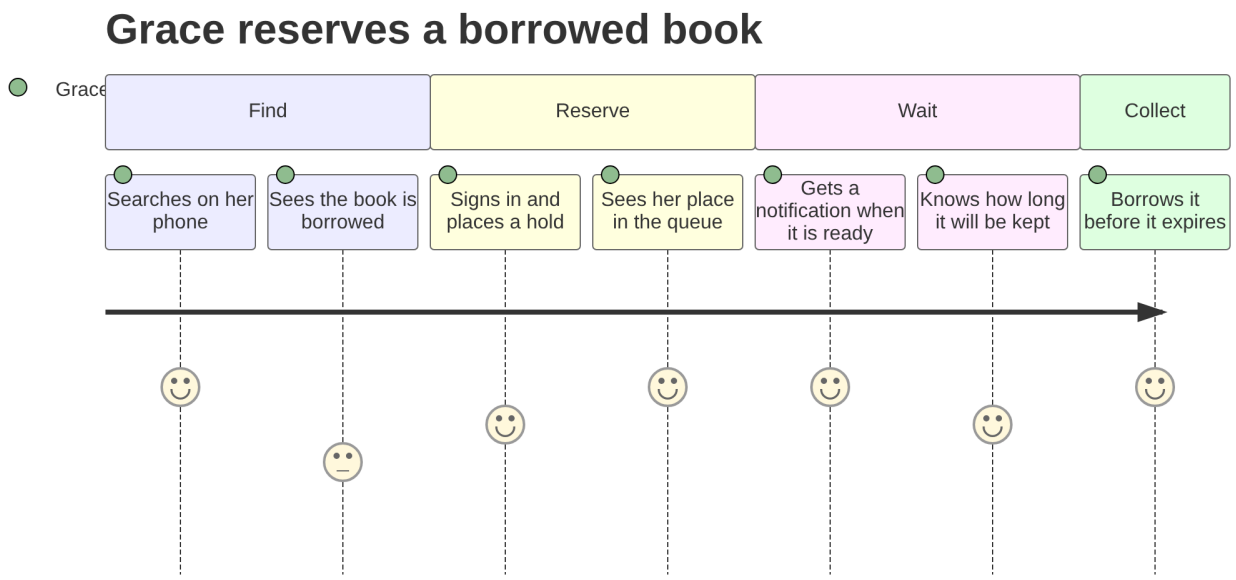
Scores run from 1 (frustrating) to 5 (delightful). Each map is followed by what the system does to remove the pain points.

Journey 1. Ava borrows at the self-service kiosk



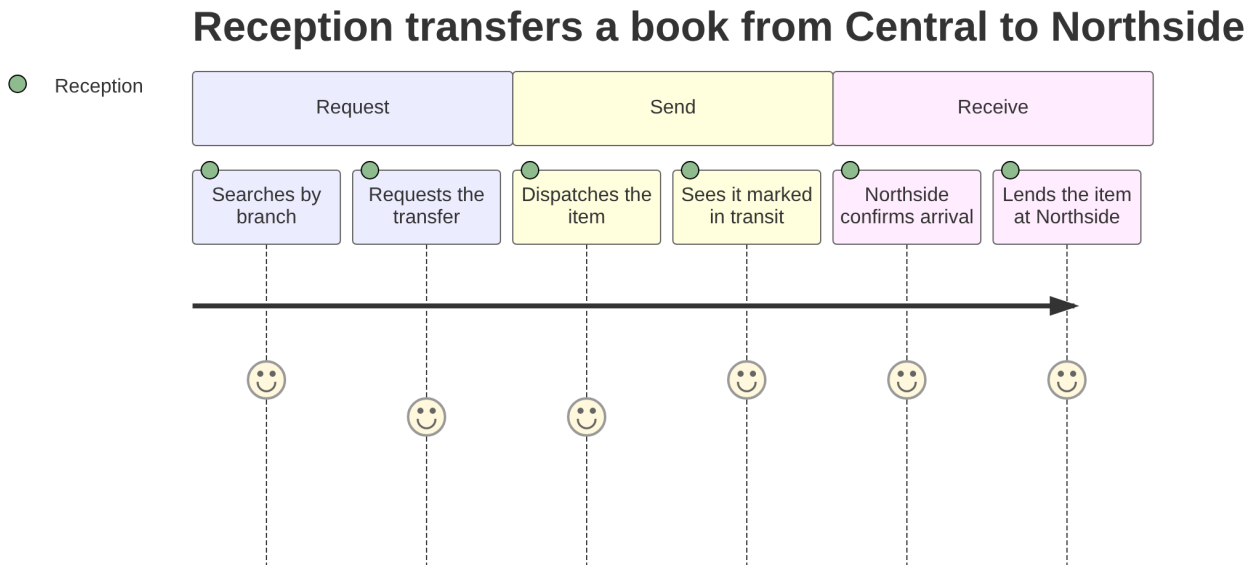
Stage	Pain point	What the system does
Arrive	Typing a PIN in public	PIN keypad, masked input, five wrong tries lock the card for five minutes
Borrow	Not knowing why an item was refused	Plain message: damaged, held for someone else, in transit, not at this branch
Leave	Forgetting to sign out	Auto sign-out after 60 idle seconds so the next person never sees the account

Journey 2. Grace waits for a book and is told when it is ready



Stage	Pain point	What the system does
Find	Not knowing if it will ever be free	Item page shows status and how many are waiting
Wait	Missing the call	Email, SMS or both according to her preference; item kept for three days
Collect	Someone else taking it	While a hold is Ready nobody else can borrow the item

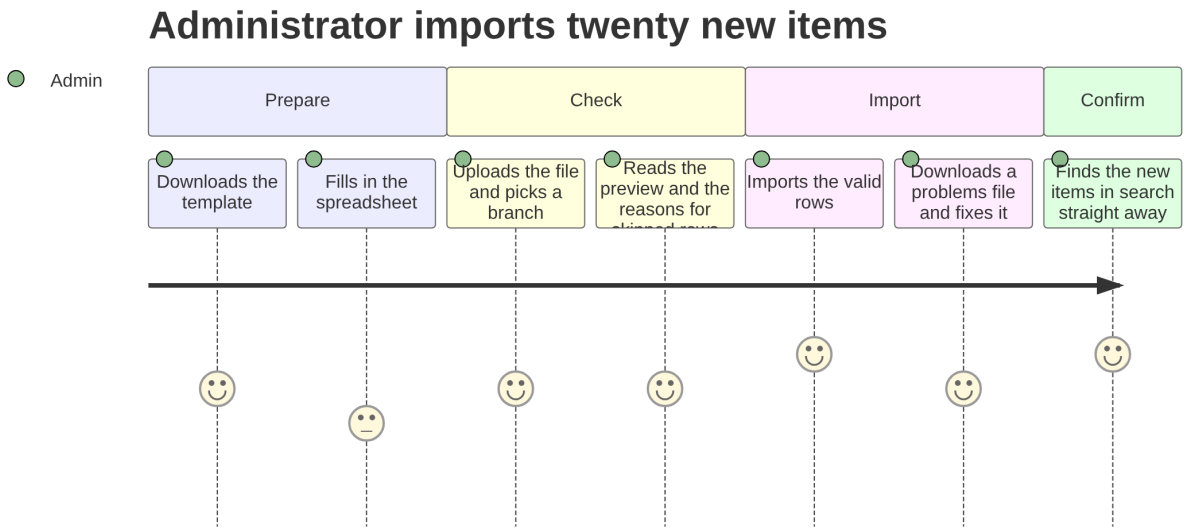
Journey 3. Reception moves a book to another branch



Stage	Pain point	What the system does
Send	Item lent while on a truck	In-transit items cannot be lent

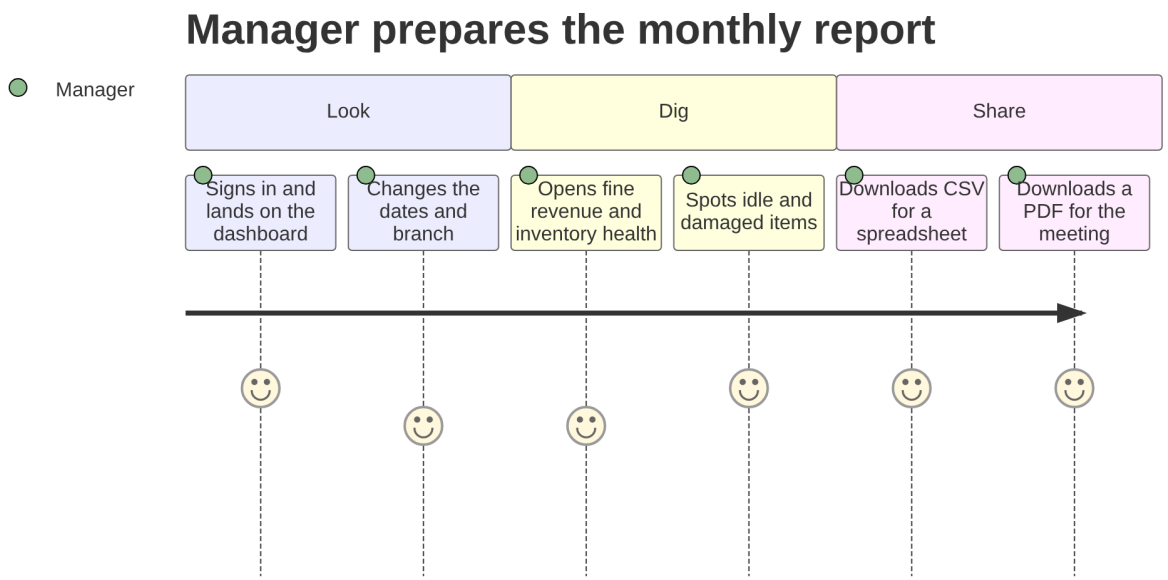
Stage	Pain point	What the system does
Receive	Nobody knows where it is	Item changes branch only when the receiving branch confirms; both branches see the same state

Journey 4. An administrator imports new stock



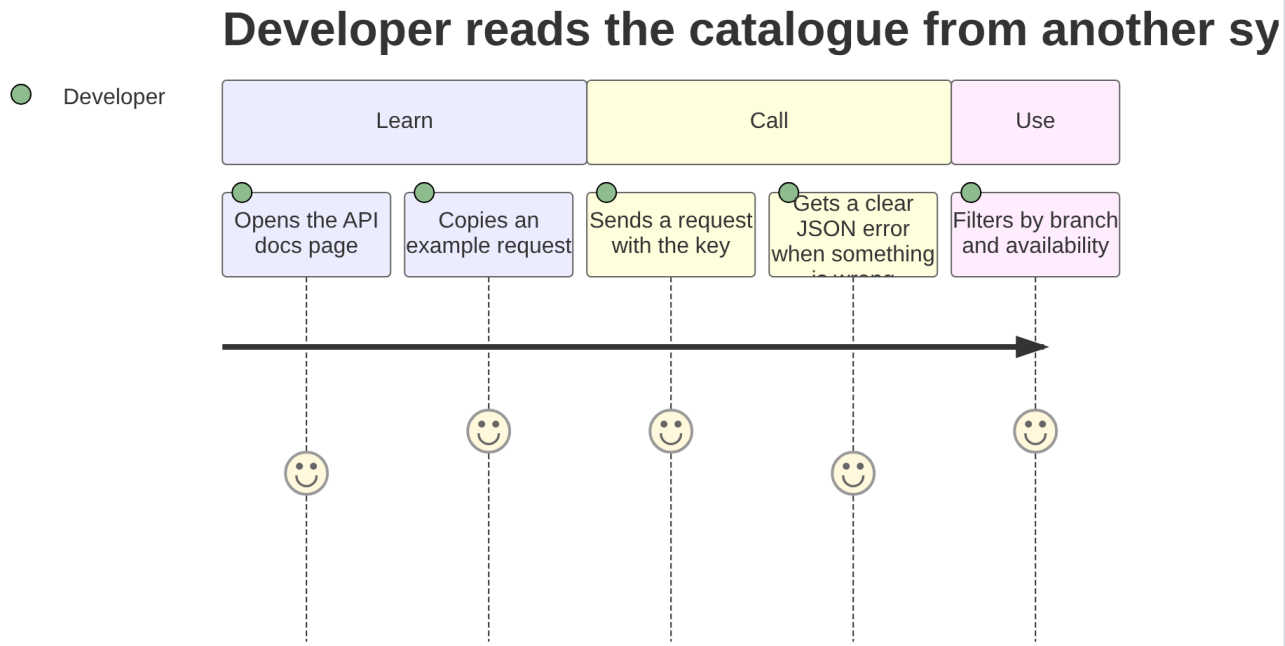
Stage	Pain point	What the system does
Check	Bad data silently entering the catalogue	Every row validated; nothing saved until the admin confirms
Import	Having to guess what is wrong	Each skipped row has a plain reason; problems can be downloaded as a CSV and fixed

Journey 5. A manager reviews performance and exports



Stage	Pain point	What the system does
Dig	Numbers differ between screen and file	The screen, CSV and PDF are produced from the same report
Share	A PDF that is hard to read	Paginated PDF with title, filters, headings and "Page n of N"

Journey 6. A developer uses the public API



Stage	Pain point	What the system does
Learn	Guessing the parameters	Interactive docs page, OpenAPI document, `curl` and `.http` examples
Call	Unclear errors	401, 403, 404, 405, 429 each return a JSON error with a code and message

6. Requirements traceability

Client requirement	Where it is built	Evidence
3.1 Kiosk: sign in, account summary, checkout, receipt, timeout	Kiosk controller and views; `PatronAuthService`; `LendingService`	Demo item 3 (9 browser checks); 7 sign-in tests
3.2 Public read-only API with key, limits and docs	`/api/v1`, API middleware, rate limiter, OpenAPI document, docs page	Demo item 4 (10 checks); 10 catalogue tests; 429 verified
3.3 Notifications (email and SMS simulated, preference, due soon, fines, reservation)	`NotificationService`, scheduler, notifications page	Demo items 5 and 6; 7 notification tests
3.4 Multi-branch inventory and transfers	Branch on every item, staff branch context, `TransferService`, transfer screens	Demo items 2 and 7; 7 transfer tests
3.5 Reservations and waitlists	`ReservationService`, holds screens, account and kiosk	Demo item 8 (7 checks); 10 reservation tests
3.6 Data importer	`ImportService`, CSV reader, mock provider, preview and history	Demo item 9 (5 checks); 12 importer tests including the sample files
3.7 Analytics and CSV and PDF export	`ReportService`, `CsvExporter`, `PdfReportWriter`, dashboard	Demo items 10 and 11 (12 checks); 12 report tests
4 The seven integration workflows	Hooks between lending, reservations and notifications; shared catalogue; shared report document	Section 4 diagrams; demo items 3, 5, 7, 8, 9
6 Seed and reset	`SeedRunner`, `--reset-db`	Demo item 12; 13 seed tests; reset run verified
Baseline Assignment 2 rules (14-day loan, $0.50 per full late day, borrower and item CRUD, roles, public search)	Lending, inventory and borrower services	20 lending tests, 13 admin tests, 6 borrower tests
Mobile-responsive public portal	Responsive stylesheet	Phone-size checks: no sideways scrolling
Validation and useful errors	Server-side validation; plain-language messages	Importer and lending tests; friendly error pages
Admin tools not public	`[Authorize(Roles = ...)]` on every staff controller	Demo item 12 check; manager refused on admin pages

7. Test evidence

7.1 Automated tests (117 passed, 0 failed)

Area	Tests
Demo data	13
Items, categories and branches	13
Borrowers and staff	6
Lending and returning	20
Inter-branch transfers	7
Reservations and waitlists	10
Notifications	7
Patron sign-in	7
Catalogue queries	10
Data importer	12
Manager analytics and exports	12

7.2 Scripted browser run of the 12 demo items (67 checks passed, 0 failed)

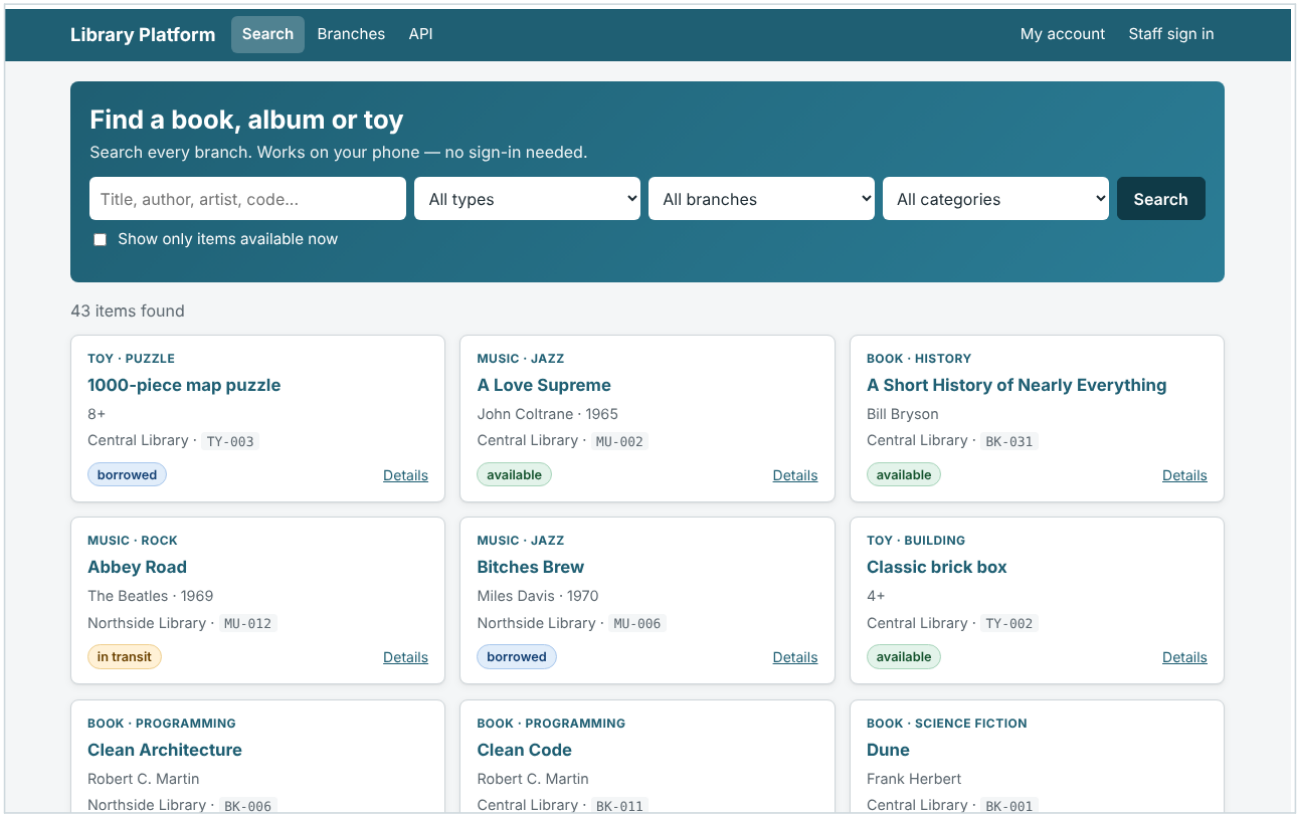
Demo item	Checks	Result
1 Fresh install and startup	3	Pass
2 Selecting a branch	3	Pass
3 Kiosk checkout and account summary	9	Pass
4 Public API requests	10	Pass
5 Borrowing and notification logging	2 (plus kiosk loans in item 3)	Pass
6 Due-date and fine notification simulation	5	Pass
7 Multi-branch inventory and transfer	8	Pass
8 Reservation and next-person notification	7	Pass
9 CSV or external import	5	Pass
10 and 11 Manager analytics and exports	12	Pass
12 Seeding and reset	13 seed tests; `--reset-db --exit` run	Pass
Access control and mobile layout	3	Pass

7.3 Other checks

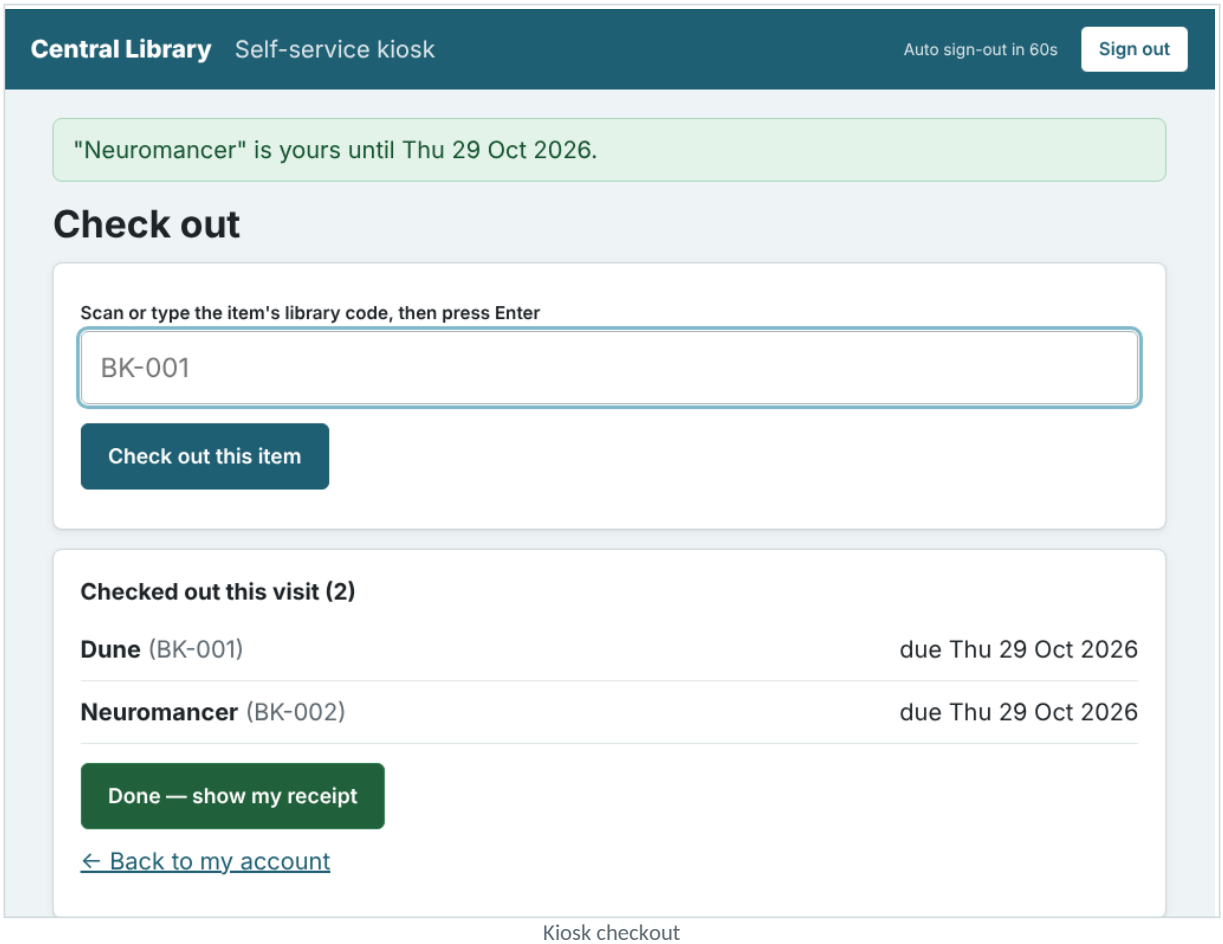
- Fresh copy of the repository with an empty data folder: builds with no package download, creates the demo data on first start, serves the site and the API.
- Release build published and run from the publish folder: pages, stylesheet and API work.
- API rate limit: the 61st request in a minute returns 429 with a `Retry-After` header.
- PDF exports rendered to images and read page by page; CSV exports open with a byte-order mark and neutralise formula characters.

Where this was run: a clean Linux environment with the .NET 10 SDK and a headless Chromium browser. It has not yet been run on the assessor's own virtual machine image.

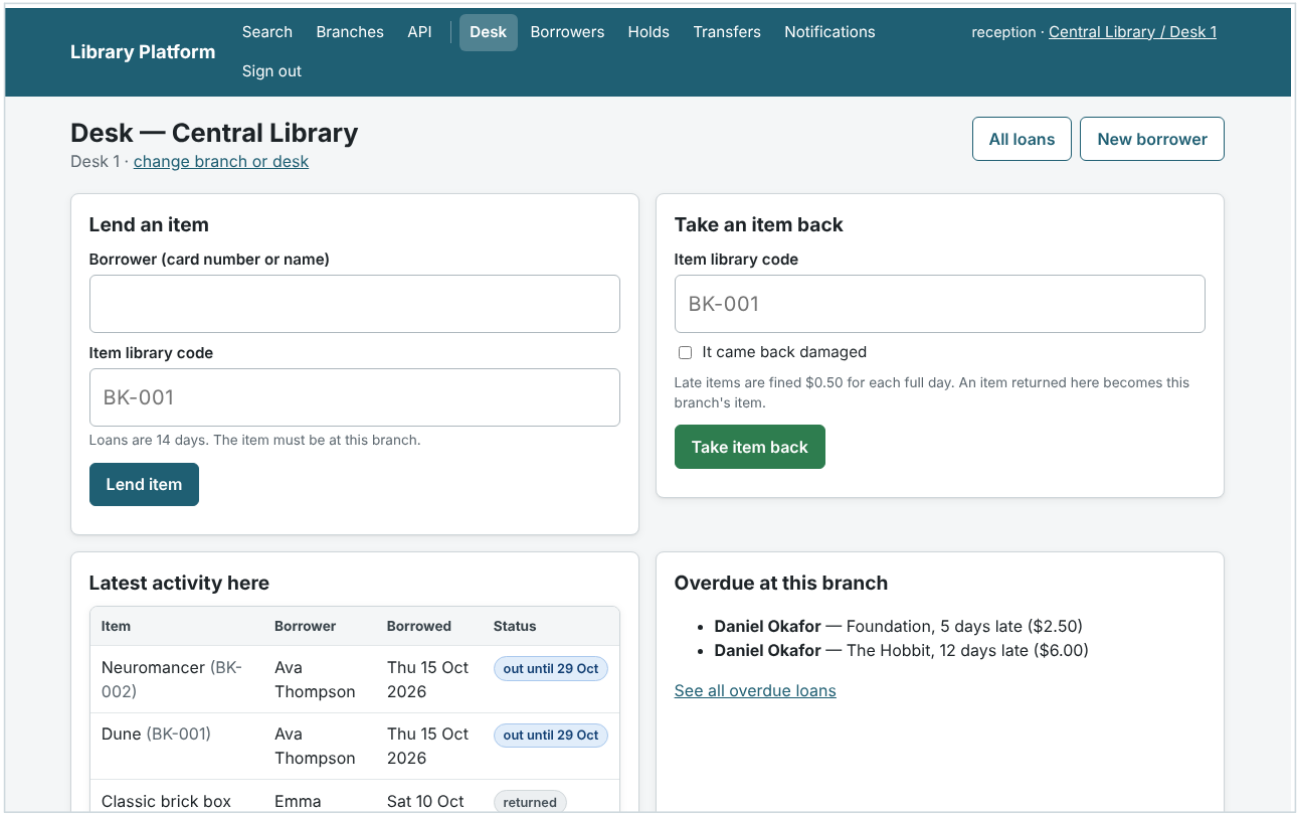
8. Screens



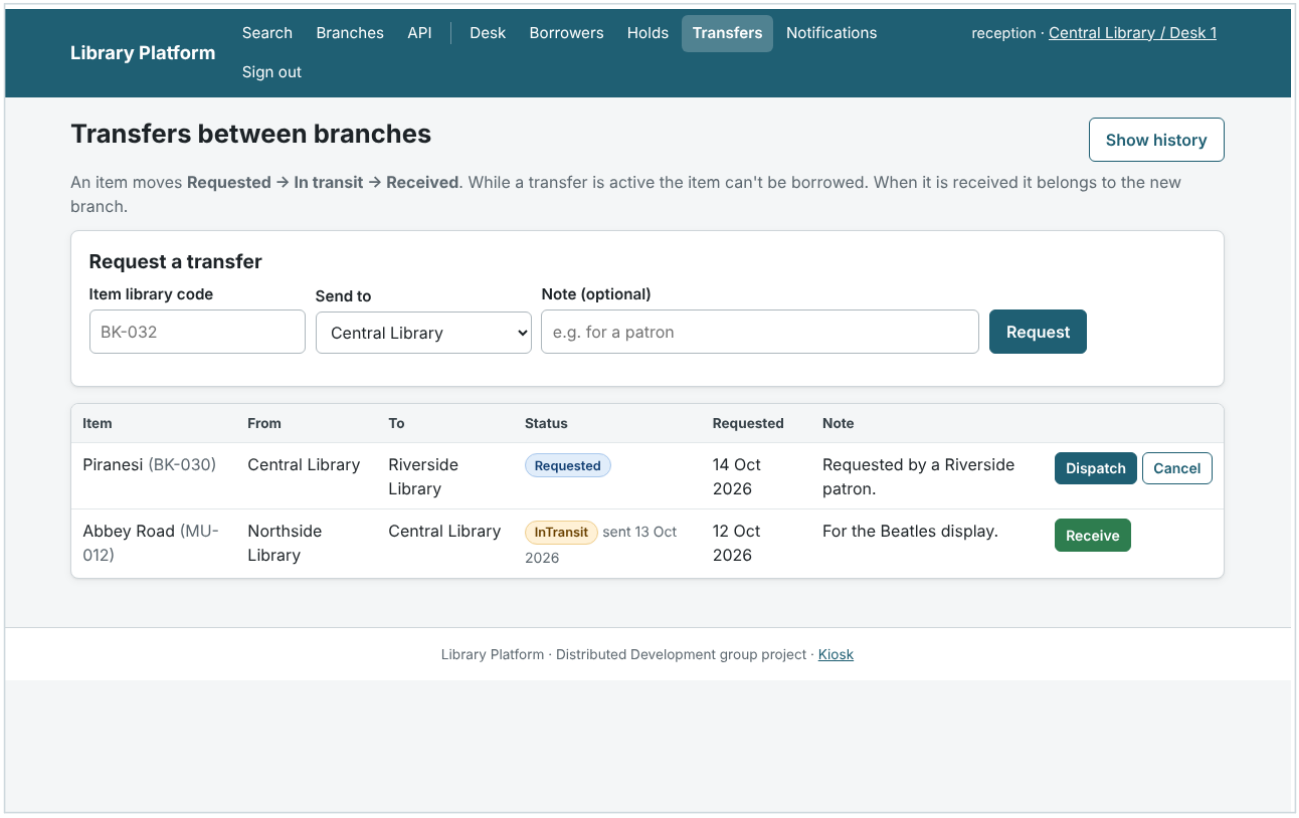
Public search



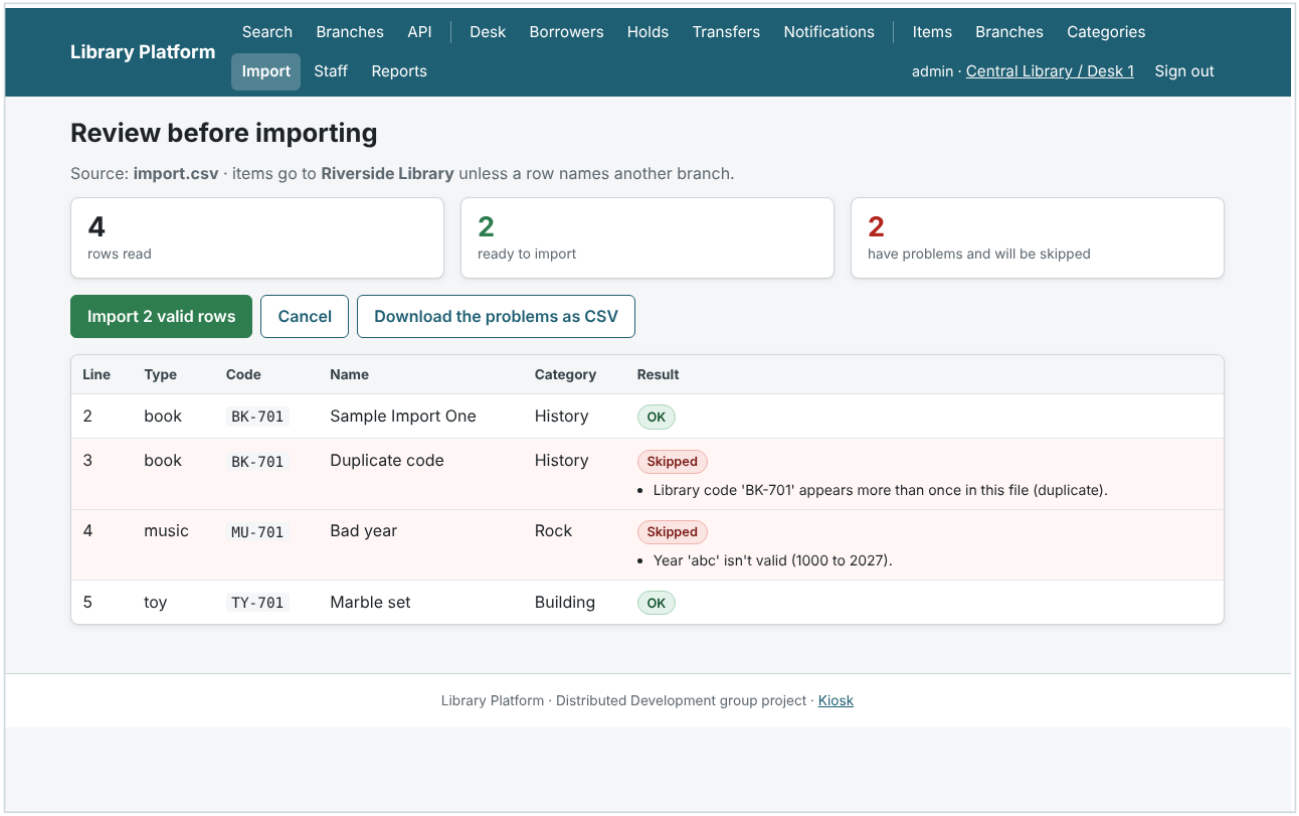
Kiosk checkout



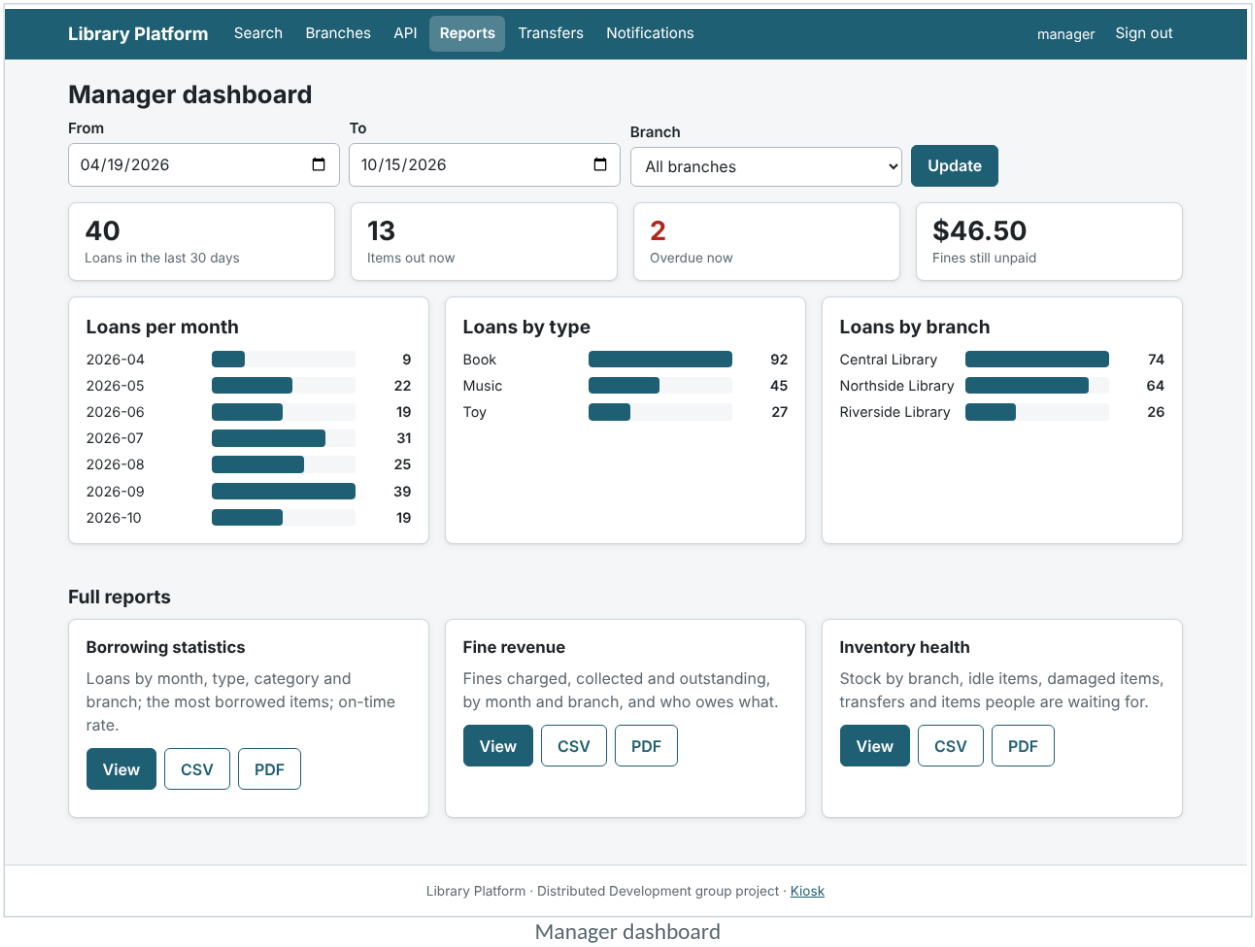
Desk



Transfers



Import preview



Manager dashboard

9. Known limitations

- JSON file storage suits a demo-sized library, not a very large one.
- Notifications are simulated; nothing is sent. The daily check runs once a day, or when an administrator presses the button.
- One shared demo API key; the site runs on plain HTTP for the demo.
- Holds are per copy, not per title.
- Fines are recorded and reported but there is no payment gateway.
- Tested on Linux in a clean environment; to be re-run on the assessor's virtual machine.